

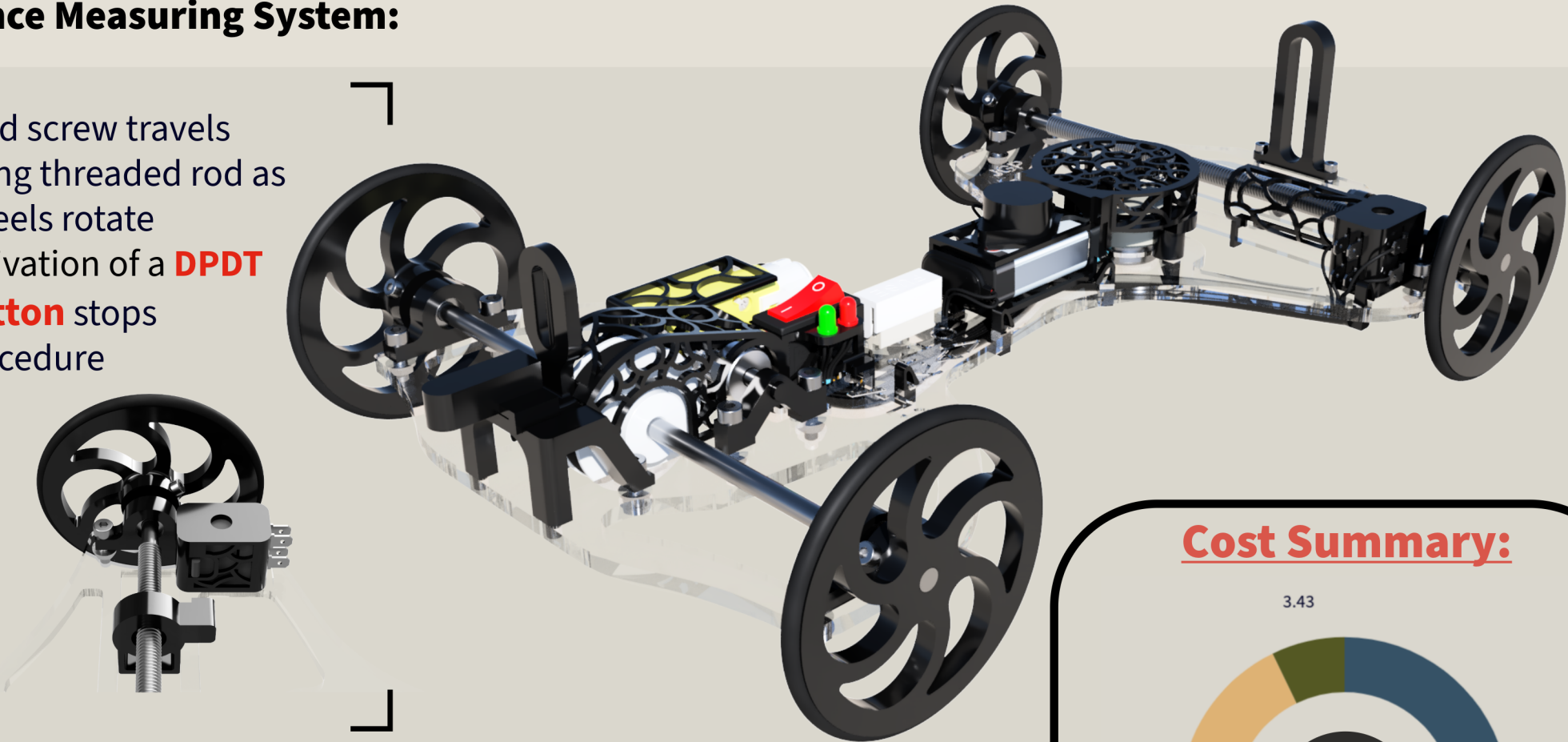


Brief:

To create a repeatable device which makes contact with the wall hitting a vertical target, remaining stationary for **10-15s**, before moving back to its starting position on a horizontal target.

Distance Measuring System:

- Lead screw travels along threaded rod as wheels rotate
- Activation of a **DPDT button** stops procedure



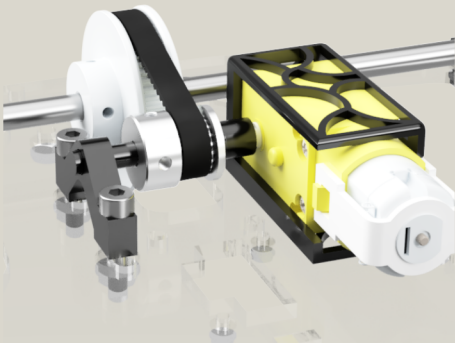
Timing System:

- Motor runs for **13.5s** per half revolution
- A **loaded spring** guarantees switch is activated



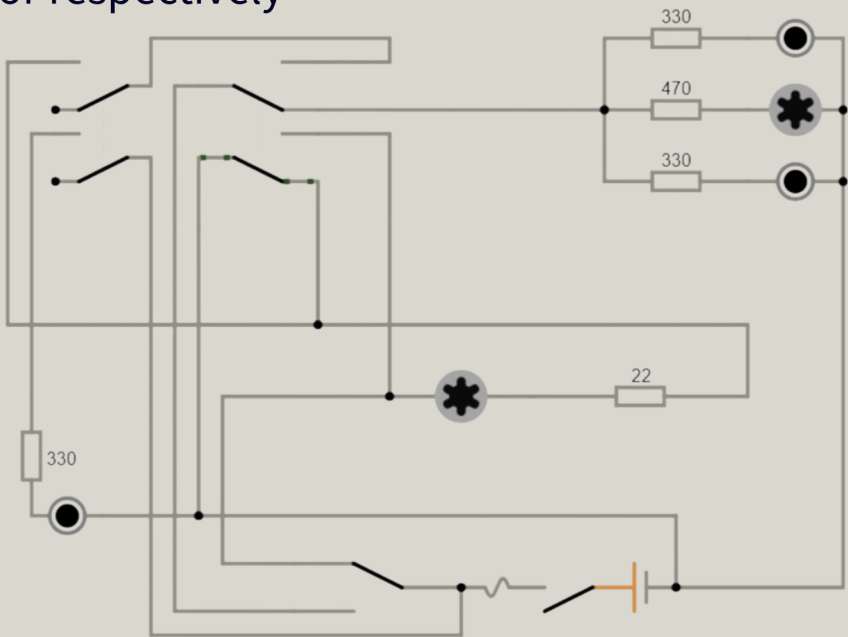
Motor Driven Pulley System:

- Gear ratio of **1:3**
- Front axle drive prevents slip affecting distance measurement

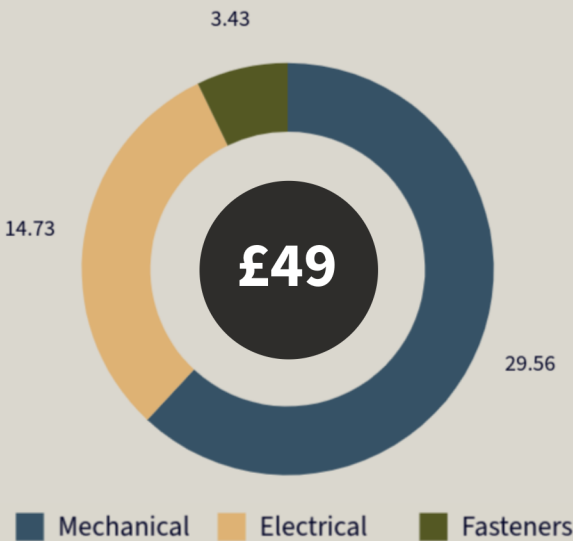


The Electrical Circuit:

- Voltage and current are **limited by resistors** to slow motors to increase control and ensure safety
- A **470Ω** and **22Ω** resistor are used for the timing & main motor respectively



Cost Summary:



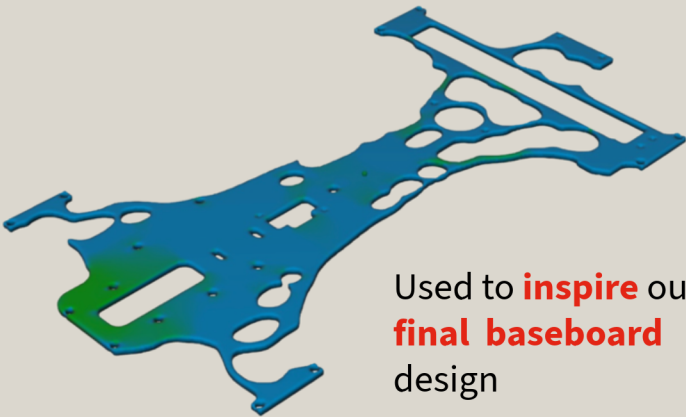
Three most expensive components:

- Pulleys and Belt = **£3.83**
- Main Motor (6V DC) = **£3.14**
- Timer Motor = **£2.33**

Total BoM Cost = £49.14
IMechE BoM Cost = £44.91

IMechE BoM - doesn't take into consideration parts under 20p

Chassis FEA and Generative Design:



Used to **inspire** our **final baseboard design**